

Task 1

CJD

2025-12-01

Librarys:

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.1      v stringr   1.5.2
## v ggplot2    4.0.0      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.1.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
```

Set WD and import data

```
# CHANGE THIS TO YOUR OWN WD
setwd("/Users/charliedicker/Documents/Uni/Year 1/R basics (data science)/Coursework/Exported_data")

gdp_per_capita_df <- read.csv("gdp_per_capita_df.csv")
```

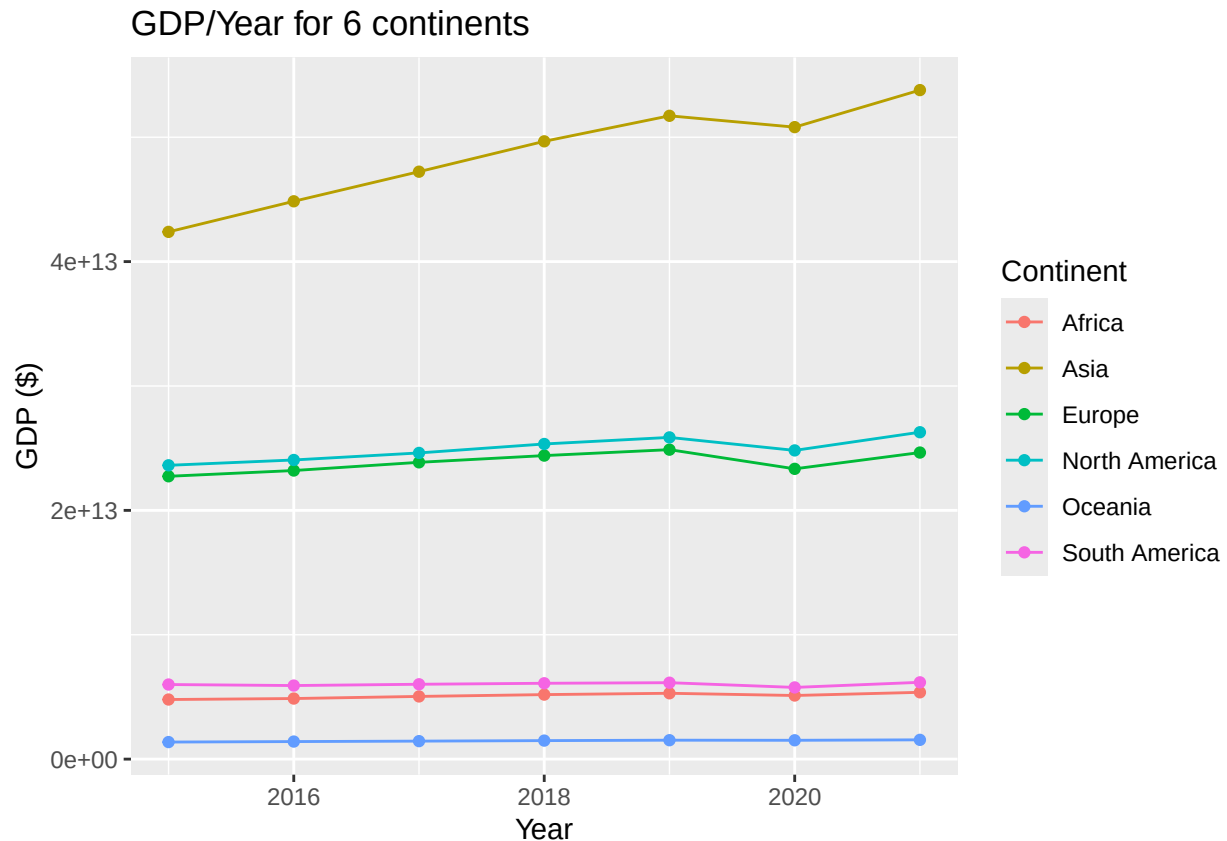
Calculate the total GDP of each continent

```
continent_gdp_totals <- gdp_per_capita_df %>%
  group_by(Continent, Year) %>%
  summarise(
    Total_GDP = sum(Gdp, na.rm = TRUE)
  ) %>%
  ungroup()
```

```
## 'summarise()' has grouped output by 'Continent'. You can override using the
## '.groups' argument.
```

Plot a graph of GDP for each continent, vs time

```
continent_gdp_totals %>% ggplot(aes(x = Year, y = Total_GDP, colour = Continent)) +
  geom_point() +
  geom_line() +
  labs(title = "GDP/Year for 6 continents", y = "GDP ($)")
```



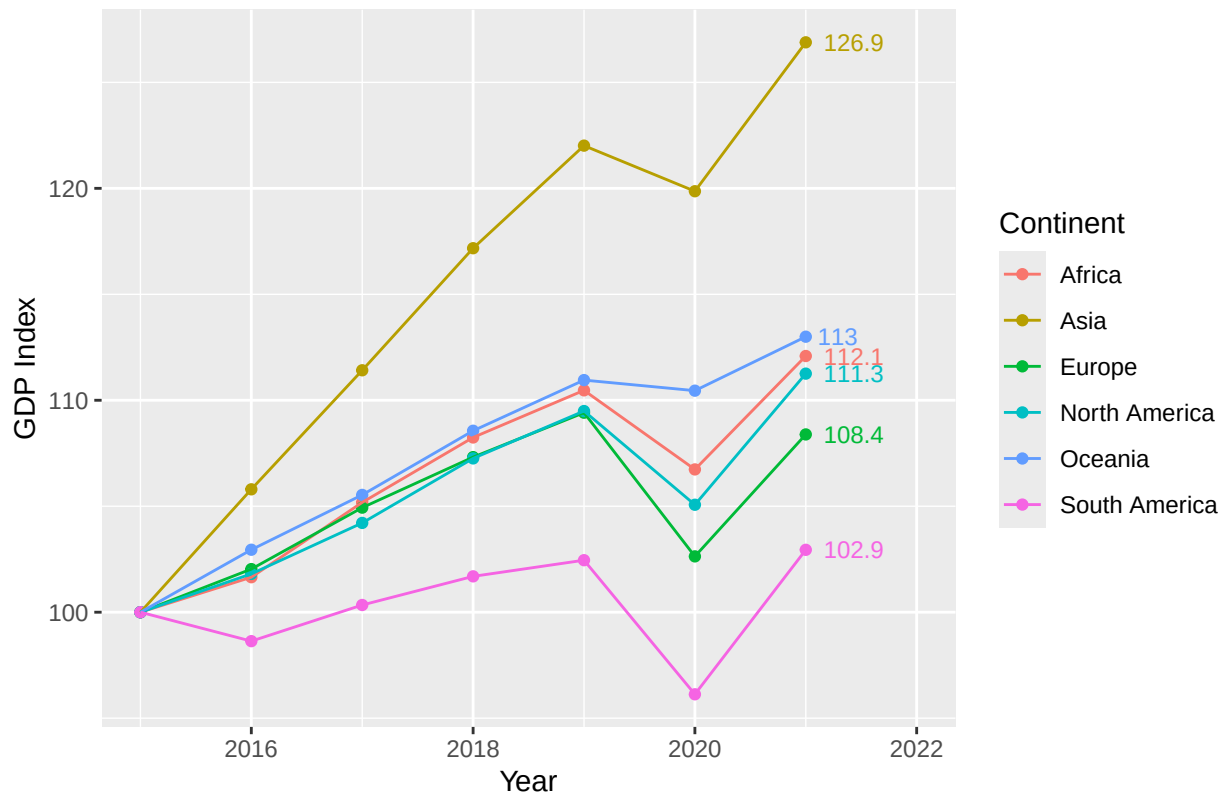
I decided this graph could show more if they all started at an index of 100

```
continent_gdp_totals <- continent_gdp_totals %>%
  group_by(Continent) %>%
  mutate(
    GDP_Index = Total_GDP / first(Total_GDP) * 100
  )
```

We now replot the graph, using normalised values

```
continent_gdp_totals %>% ggplot(aes(x = Year, y = GDP_Index, colour = Continent)) +
  geom_point() +
  geom_line() +
  geom_text(
    data = continent_gdp_totals %>% filter(Year == 2021),
    aes(label = round(GDP_Index, 1)),
    hjust = -0.3,
    size = 3,
    show.legend = FALSE) +
  expand_limits(x = 2022) +
  labs(title = "GDP Index/Year for 6 continents", y = "GDP Index")
```

GDP Index/Year for 6 continents



Calculate the weighted and unweighted growth of each continent

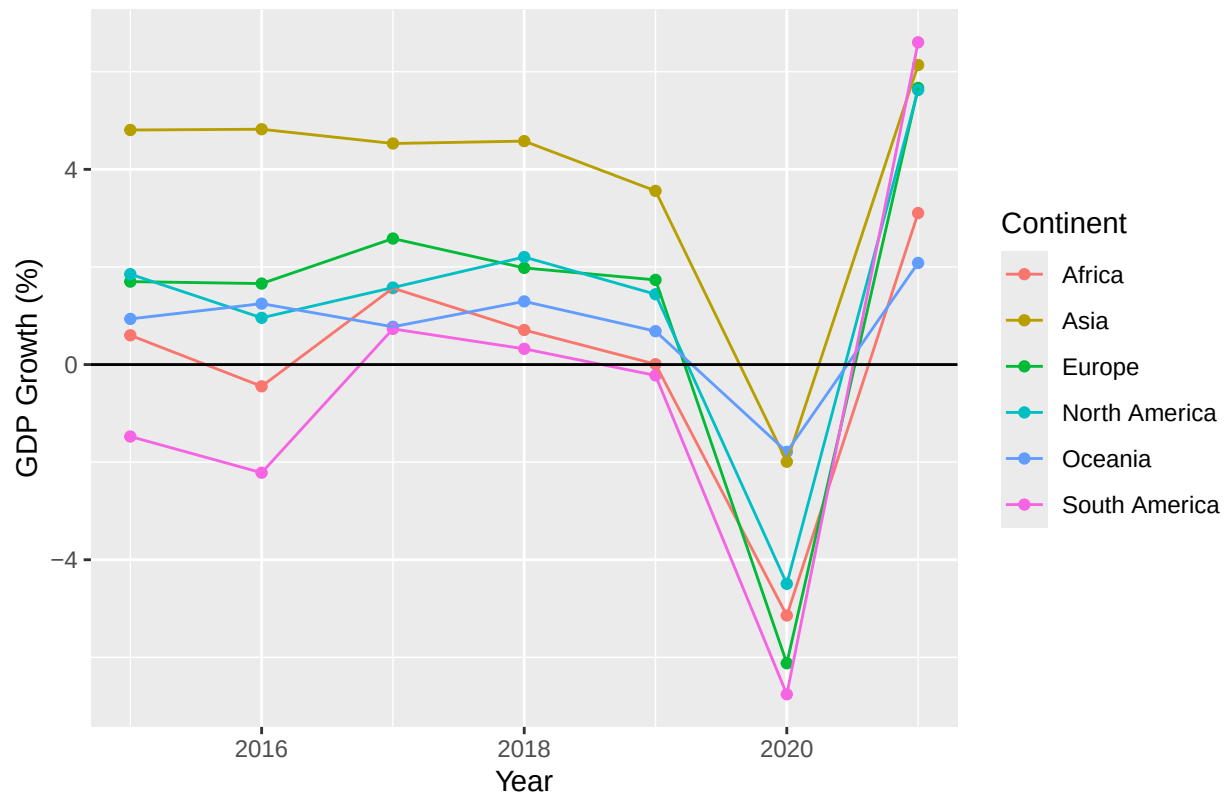
```
continents_growth_df <- gdp_per_capita_df %>%
  group_by(Continent, Year) %>%
  summarise(
    Weighted_Growth = sum(Growth * Gdp, na.rm = TRUE) / sum(Gdp, na.rm = TRUE),
    Avg_Growth = mean(Growth, na.rm = TRUE)
  ) %>%
  ungroup()
```

'summarise()' has grouped output by 'Continent'. You can override using the
'.groups' argument.

Plot the growth graphs of each continent

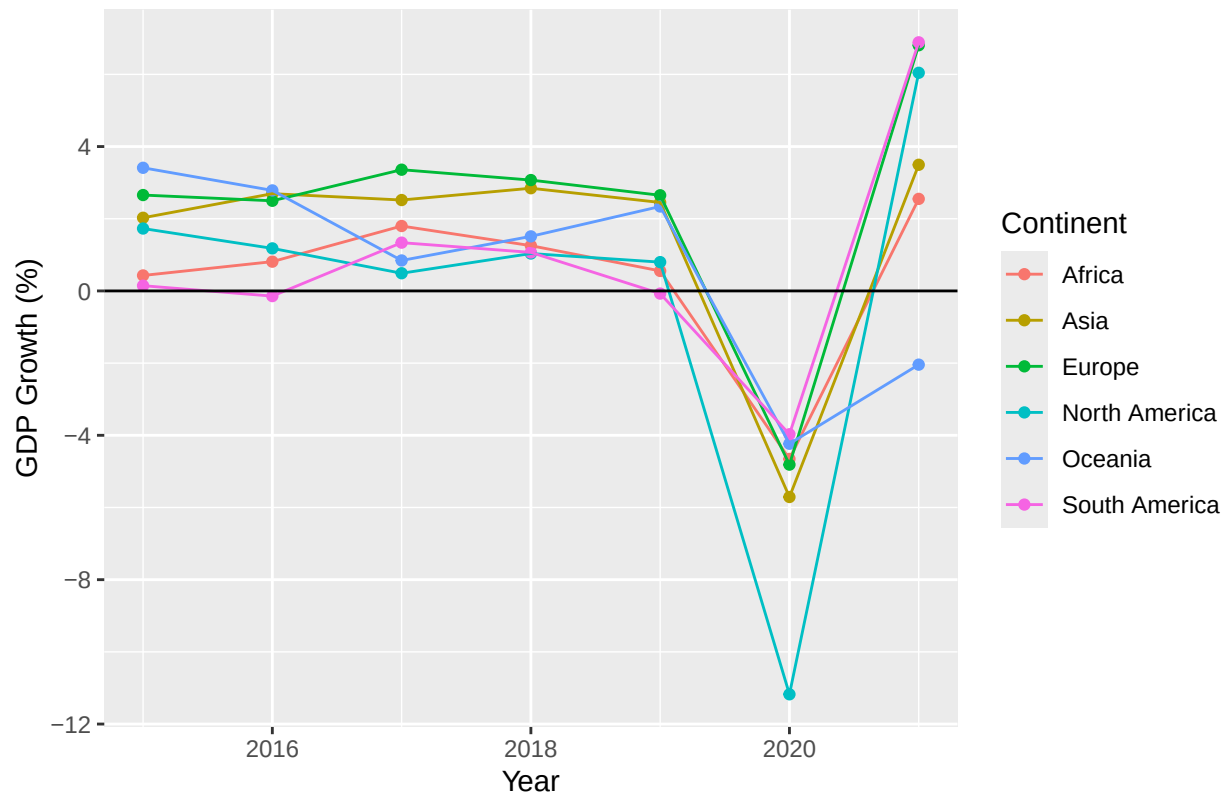
```
continents_growth_df %>% ggplot(aes(x = Year, y = Weighted_Growth, colour = Continent)) +
  geom_point() +
  geom_line() +
  geom_hline(yintercept = 0) +
  labs(title = "GDP Growth (weighted)/Year for 6 continents", y = "GDP Growth (%)")
```

GDP Growth (weighted)/Year for 6 continents



```
continents_growth_df %>% ggplot(aes(x = Year, y = Avg_Growth, colour = Continent)) +
  geom_point() +
  geom_line() +
  geom_hline(yintercept = 0) +
  labs(title = "GDP Growth (avg)/Year for 6 continents", y = "GDP Growth (%)")
```

GDP Growth (avg)/Year for 6 continents

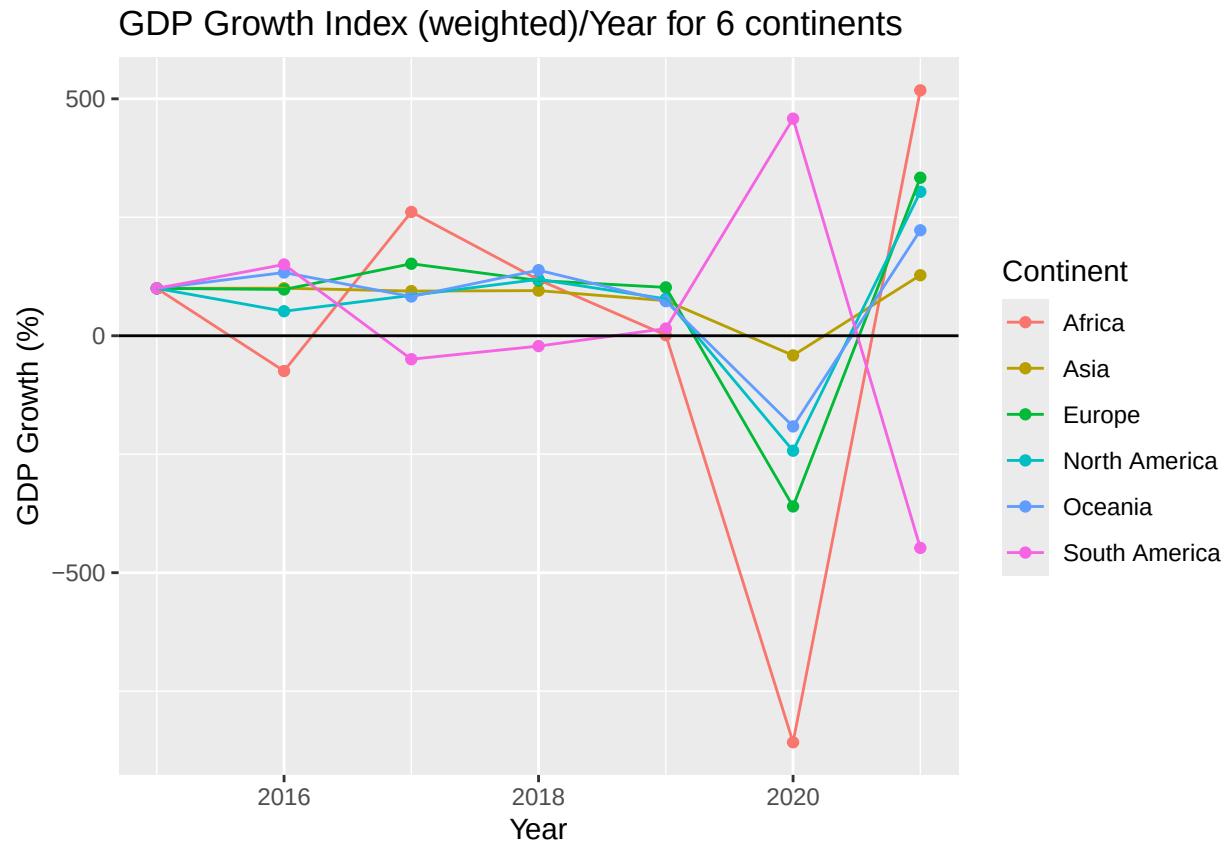


I decided to do a similar thing with growth about index

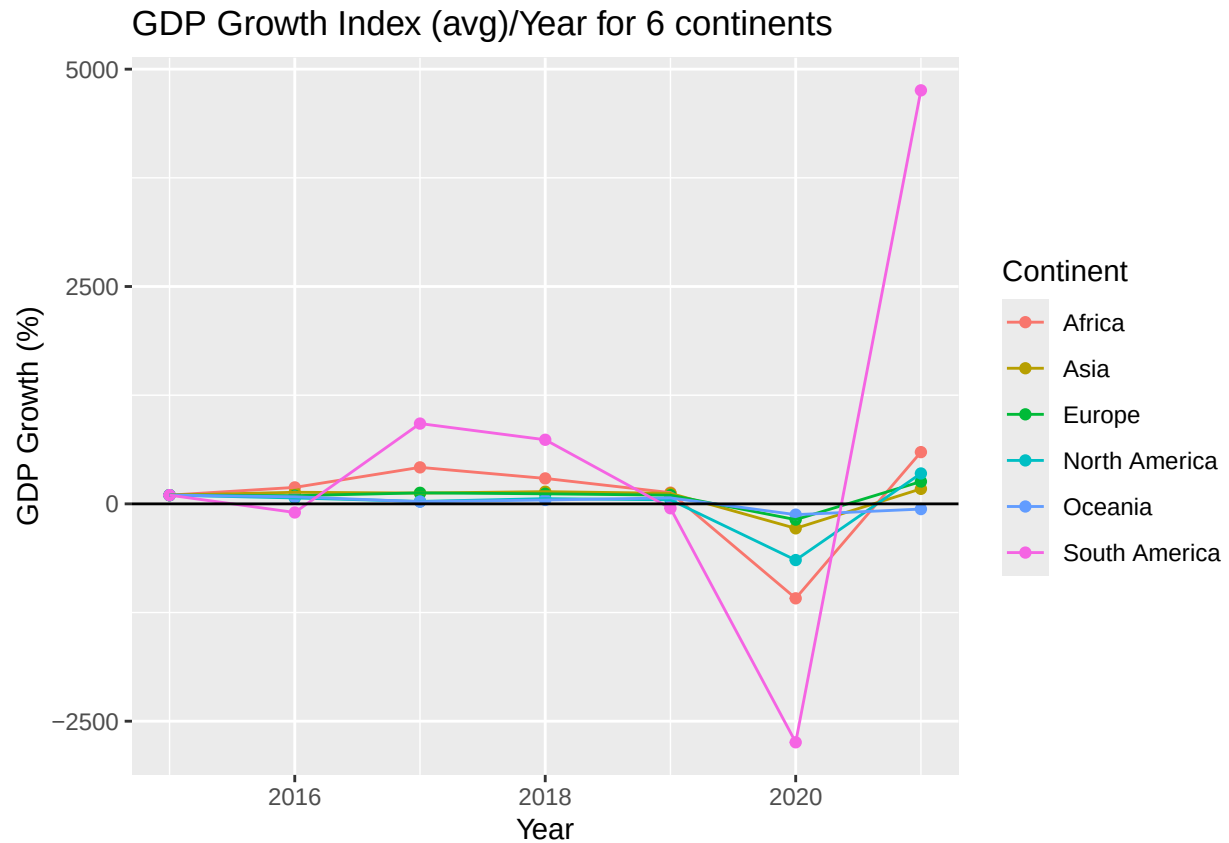
```
continents_growth_df <- continents_growth_df %>%
  group_by(Continent) %>%
  mutate(
    Avg_Growth_Index = Avg_Growth / first(Avg_Growth) * 100,
    Weighted_Growth_Index = Weighted_Growth / first(Weighted_Growth) * 100
  )
```

Plotting the graphs

```
continents_growth_df %>% ggplot(aes(x = Year, y = Weighted_Growth_Index, colour = Continent)) +
  geom_point() +
  geom_line() +
  geom_hline(yintercept = 0) +
  labs(title = "GDP Growth Index (weighted)/Year for 6 continents", y = "GDP Growth (%)")
```



```
continents_growth_df %>% ggplot(aes(x = Year, y = Avg_Growth_Index, colour = Continent)) +
  geom_point() +
  geom_line() +
  geom_hline(yintercept = 0) +
  labs(title = "GDP Growth Index (avg)/Year for 6 continents", y = "GDP Growth (%)")
```



REF:

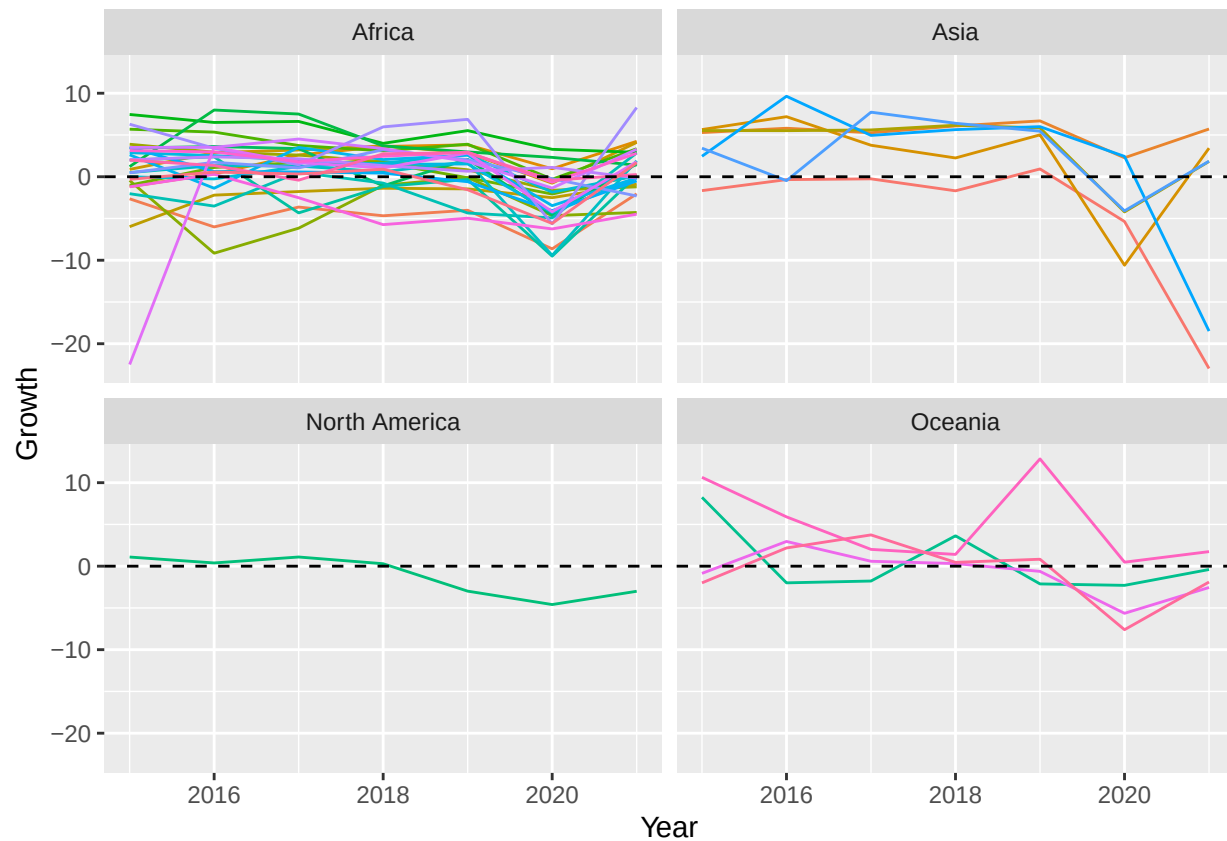
28/11/2025 - asked ChatGPT - "i have text near the last points on the x axis, how can i make it so it doesn't overlap the axis"

Plot ldc growth, each separate country in a certain continent (6 graphs) Create a ldc database

```
ldc <- gdp_per_capita_df %>%
  filter (Ldc == TRUE)
```

Plot the graphs

```
ggplot(ldc, aes(x = Year, y = Growth, color = Entity)) +
  geom_line() +
  facet_wrap(~ Continent) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  theme(legend.position = "none")
```

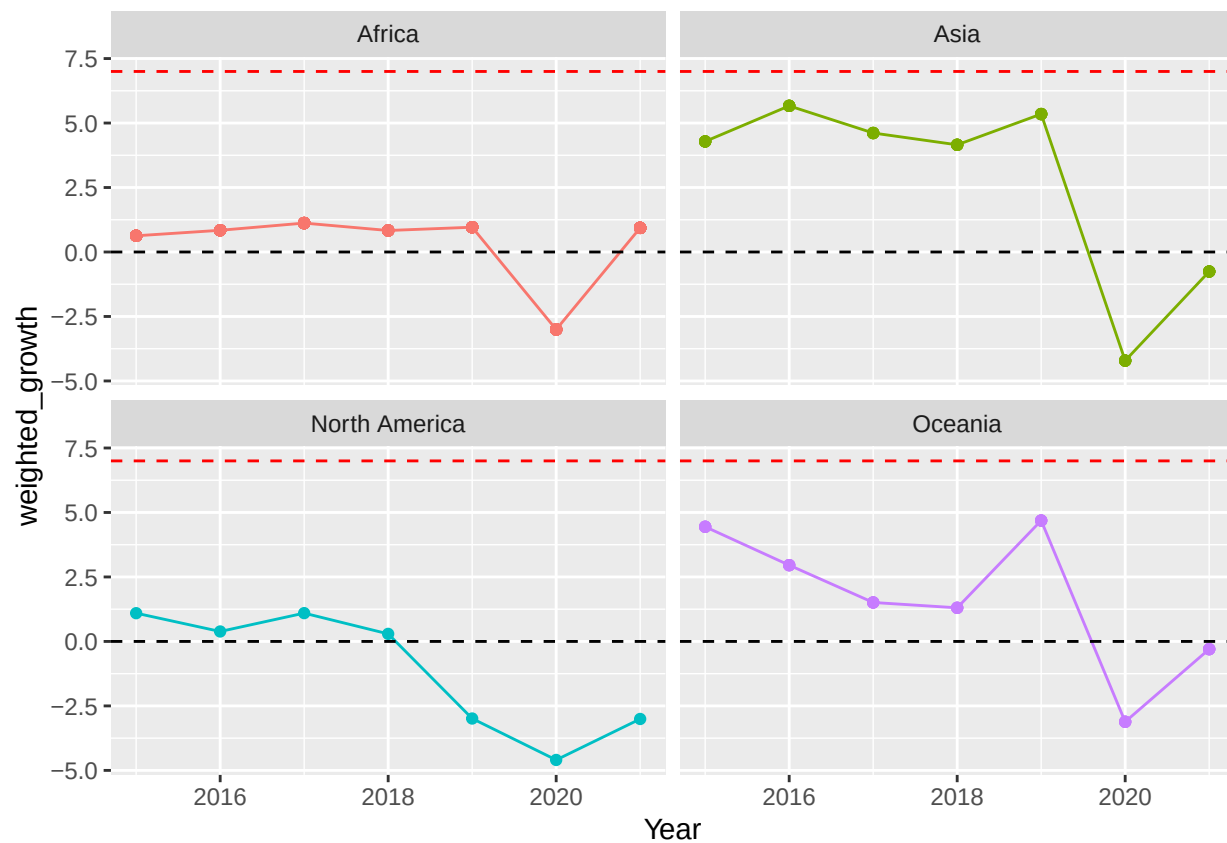


Waiiiittttt on the y axis what exactly should i put? gdp, gdp per capita or growth???

7 do a weighted average of ldc growth rate for each contine, then plot a graph of each continent and how they compare

```
ldc_weighted <- ldc %>%
  group_by(Year, Continent) %>%
  mutate(
    weighted_growth = sum(Gdp_per_capita * Growth, na.rm = TRUE) / sum(Gdp_per_capita, na.rm = TRUE)
  ) %>%
  ungroup()

ggplot(ldc_weighted, aes(x = Year, y = weighted_growth, colour = Continent)) +
  geom_line() +
  geom_point() +
  facet_wrap(~ Continent) +
  theme(legend.position = "none") +
  geom_hline(yintercept = 7, linetype = "dashed", color = "red") +
  geom_hline(yintercept = 0, linetype = "dashed", color = "black")
```



REF:

28/11/2025 - asked ChatGPT - “I have a facet plot in ggplot with some continents having tons of lines and some only one. How can I adjust the y-axis scales so each graph uses the right space for its lines and they don’t look squished or empty?”